7B INTERMOLECULAR INTERACTIONS AND PHYSICAL PROPERTIES

SPECIFICATION REFERENCE 7.3 7.5 7.6

LEARNING OBJECTIVES

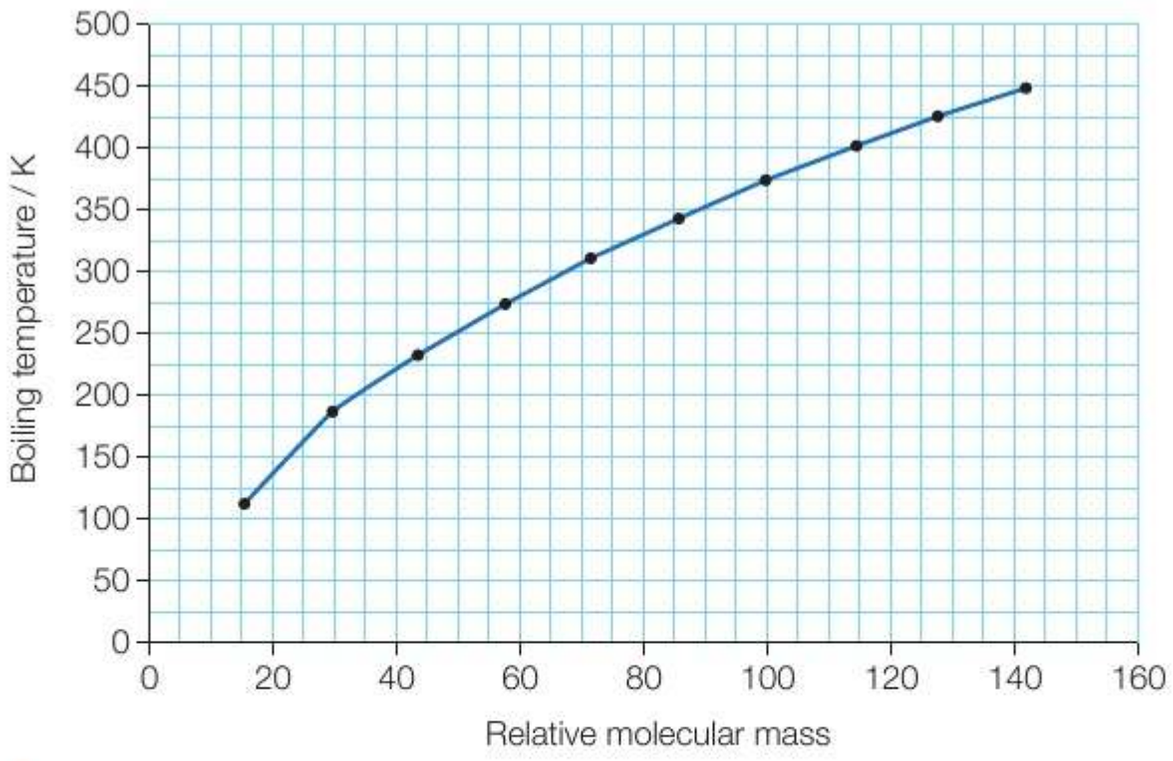
- Understand in terms of intermolecular interactions, the physical properties shown by materials, including:
  (i) the trends in boiling temperatures of alkanes with increasing chain length
  (ii) the effect of branching in the carbon chain on the boiling temperatures of alkanes
  (iii) the relatively low volatility (higher boiling temperatures) of alcohols compared with alkanes with a similar number of electrons
  (iv) the trends in boiling temperatures of the hydrogen halides HF to HI.
- Understand the following anomalous properties of water resulting from hydrogen bonding:
  (i) its relatively high melting and boiling temperatures
  (ii) the density of ice compared with that of water.
- Understand the reasons for the choice of solvents, including:
  (i) water, to dissolve some ionic compounds, in terms of the hydration of the ions
  (ii) water, to dissolve simple alcohols, in terms of hydrogen bonding
  (iii) water, as a poor solvent for some compounds, in terms of inability to form strong hydrogen bonds
  (iv) non-aqueous solvents, for compounds which have similar intermolecular interactions to those in the solvent.

BOILING TEMPERATURES OF ALKANES AND ALCOHOLS

UNBRANCHED ALKANES

The alkanes are a homologous series of hydrocarbons with the general formula C_nH_{2n+2} (see **Topic 4**).

The graph in **fig A** shows the relationship between the boiling temperature and the relative molecular mass for the first 10 unbranched alkanes (i.e. CH_4 to $C_{10}H_{22}$ inclusive).



▲ **fig A** The relationship between boiling temperature and relative molecular mass for the first ten alkanes.

EXAM HINT

Think about how important these intermolecular forces are in explaining the properties of polymers.

The only significant intermolecular interaction between alkane molecules is the London force.

There are two reasons for the increase in boiling temperature with increasing molecular mass:

1 As molecular mass increases, the number of electrons per molecule increases and so the instantaneous and induced dipoles also increase (see **Topic 7A.1**).
2 As the length of the carbon chain increases, the number of points of contact between adjacent molecules increases. Instantaneous dipole-induced dipole forces exist at each point of contact between the molecules, so the more points of contact there are, the greater the overall intermolecular (London) force of attraction.

You can see the relationship between chain length and points of contact using the skeletal formulae of the alkanes (**fig B**).



▲ **fig B** Skeletal formulae of propane, butane and pentane

Because of their shapes, the molecules of the alkanes fit together very well and pack very closely. There are points of contact all the way along the chain. The longer the chain, the more points of contact there are.

BRANCHED ALKANES

Branched chain alkanes have lower boiling temperatures than their unbranched isomers (**table A**).

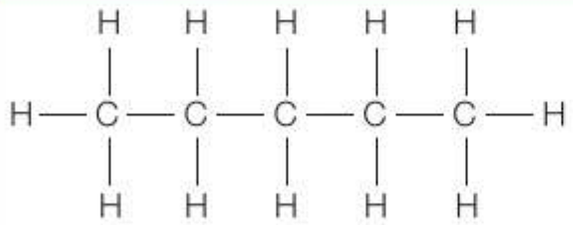
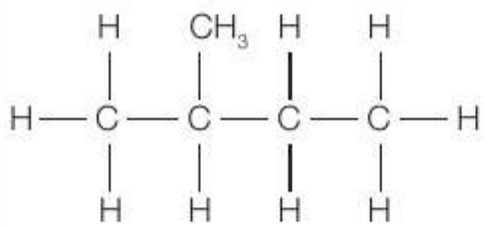
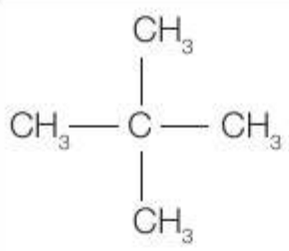
NAME OF ALKANE	STRUCTURAL FORMULA	BOILING TEMPERATURE / K
pentane		309
2-methylbutane		301
2,2-dimethylpropane		283

table A

The more branching in the molecule, the fewer points of contact between adjacent molecules; i.e. they do not pack together as well. This leads to a decrease in the overall intermolecular force of attraction between molecules and a decrease in boiling temperature.

ALCOHOLS

Alcohols are a homologous series of compounds with the general formula $C_nH_{2n+1}OH$ (see **Topic 10**). They contain an –O—H group and can therefore form intermolecular hydrogen bonds in addition to London forces. This additional bonding affects the boiling temperature of alcohols when compared with the equivalent alkane. This is shown in **table B**.

FORMULA OF ALCOHOL	NUMBER OF ELECTRONS IN MOLECULE	BOILING TEMPERATURE / K	FORMULA OF ALKANE	NUMBER OF ELECTRONS IN MOLECULE	BOILING TEMPERATURE / K
CH_3OH	18	338	CH_3CH_3	18	184
CH_3CH_2OH	26	352	$CH_3CH_2CH_3$	26	231
$CH_3CH_2CH_2OH$	34	370	$CH_3CH_2CH_2CH_3$	34	267
$CH_3CH_2CH_2CH_2OH$	42	390	$CH_3CH_2CH_2CH_2CH_3$	42	309

table B Boiling temperatures of alcohols and alkanes.

Consider the case of methanol (CH_3OH) and ethane (CH_3CH_3). Both molecules have a similar chain length. They also have the same number of electrons. If the intermolecular interactions in each were only London forces, then their boiling points would be almost identical. However, the boiling point of the alcohol is higher than that of the alkane. This is because of the hydrogen bonding that exists

between methanol molecules. This does not exist between molecules of ethane. The additional force of attraction increases the energy required to separate the molecules.

You will notice the same trend in the other compounds in **table B**.

It is sometimes stated that the predominant, or main, bonding in alcohols is hydrogen bonding. We have already mentioned that this is not always the case (**Topic 7A.1**). **Table C** provides evidence that for the first few members of the alcohol series hydrogen bonding is predominant. However, London forces eventually predominate as the chain length increases.

ALCOHOL	ENTHALPY CHANGE OF VAPORISATION / kJ mol^{-1}
CH_3CH_2OH	38.6
$CH_3CH_2CH_2OH$	47.5
$CH_3CH_2CH_2CH_2OH$	52.4
$CH_3CH_2CH_2CH_2CH_2OH$	57.0
$CH_3CH_2CH_2CH_2CH_2CH_2OH$	61.6

ALKANE	ENTHALPY CHANGE OF VAPORISATION / kJ mol^{-1}
$CH_3CH_2CH_3$	15.7
$CH_3CH_2CH_2CH_3$	21.0
$CH_3CH_2CH_2CH_2CH_3$	26.4
$CH_3CH_2CH_2CH_2CH_2CH_3$	31.6
$CH_3CH_2CH_2CH_2CH_2CH_2CH_3$	36.6

table C Enthalpy changes of vaporisation of alcohols and alkanes.

The enthalpy change of vaporisation is a measure of the amount of energy that is required to *completely* separate the molecules of a liquid and convert it into a gas at the same temperature. It is, therefore, a direct measure of the strength of the intermolecular interactions. The greater the enthalpy change of vaporisation, the greater the forces of attraction between the molecules.

In the case of ethanol, the total energy required to separate one mole of molecules is 38.6 kJ.

Of this, approximately 15.7 kJ mol^{-1} can be attributed to London forces. For this reason, the hydrogen bonding is the predominant bonding, providing approximately 59% of the total.

[(38.6 – 15.7) = 22.9; 22.9 is 59% of 38.6]

You can calculate the percentage contribution for the other alcohols in a similar way. **Table D** shows the results of these calculations.

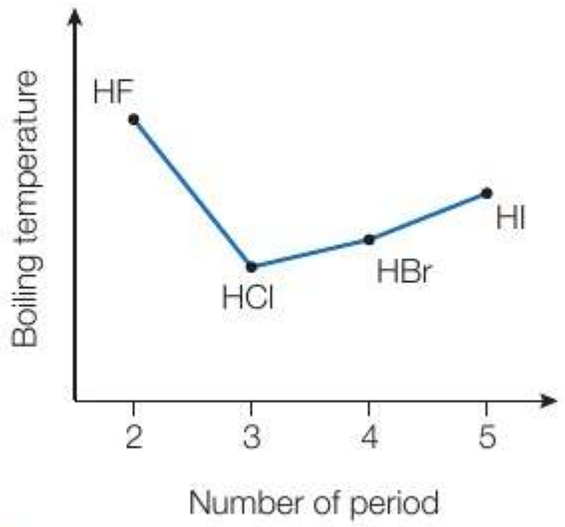
ALCOHOL	APPROXIMATE PERCENTAGE CONTRIBUTION OF HYDROGEN BONDING
$CH_3CH_2CH_2OH$	56
$CH_3CH_2CH_2CH_2OH$	50
$CH_3CH_2CH_2CH_2CH_2OH$	45
$CH_3CH_2CH_2CH_2CH_2CH_2OH$	41

table D Percentage contribution of hydrogen bonding in alcohols.

The calculations demonstrate the danger of making generalised statements such as, 'The predominant bonding in alcohols is hydrogen bonding'.

BOILING TEMPERATURES OF THE HYDROGEN HALIDES

The graph in **fig C** shows the boiling temperatures of the hydrogen halides HF to HI.



▲ **fig C** Boiling temperatures of the hydrogen halides.

The steady increase in boiling temperature from HCl to HI is the result of the increasing number of electrons per molecule, which in turn results in an increase in London forces.

EXAM HINT

You may be asked to sketch a graph or plot a graph. If you are asked to sketch, you only need to show the pattern or trend. If you are asked to plot, you need to provide clear, even scales on each axis and points plotted to within an accuracy of 1 mm^2.

DID YOU KNOW?

There are three intermolecular interactions within each compound. They are:

1 London forces
2 permanent dipole–dipole interactions
3 hydrogen bonding.

However, with HCl, HBr and HI, the electronegativity of the halogen is low enough for the permanent dipole–dipole interactions and the hydrogen bonding to be relatively weak. The predominant interaction in each compound is the London force.

The boiling temperature of HF is significantly higher than that of the other three hydrogen halides, despite having fewer electrons per molecule. The London force is weaker in HF, but the hydrogen bonding is significantly greater because of the high electronegativity of fluorine.

ANOMALOUS PROPERTIES OF WATER

Water has some anomalous properties. The following two are particularly important:

1 It has a relatively high melting and boiling temperature for a molecule with so few electrons.
2 The density of ice at 0 °C is less than that of water at 0 °C.

MELTING AND BOILING TEMPERATURES

The hydrogen bonds between water molecules are relatively strong (see **Topic 3E.2**). As a result, the overall intermolecular

forces of attraction in water are greater than would be expected from the number of electrons (10) in the molecule.

Water, therefore, has an abnormally high melting temperature (0 °C, 273 K) and boiling temperature (100 °C, 373 K) at 100 kPa pressure.

It is interesting to compare the boiling temperatures of water, ammonia and hydrogen fluoride (**table E**).

	BOILING TEMPERATURE	NUMBER OF ELECTRONS PER MOLECULE	STRENGTH OF HYDROGEN BONDING / $kJmol^{-1}$
H_2O	373 K (100°C)	10	22
NH_3	240 K (−33°C)	10	17
HF	293 K (20°C)	10	29

table E Strength of hydrogen bonding in water, ammonia and hydrogen fluoride.

The number of electrons per molecule is identical, so we would expect the London forces to be similar between each set of molecules. The differences are a result of different extents of hydrogen bonding.

The hydrogen bond strength for HF is greater than the hydrogen bond strength for H_2O, but surprisingly, its boiling temperature is lower. This is because of two factors.

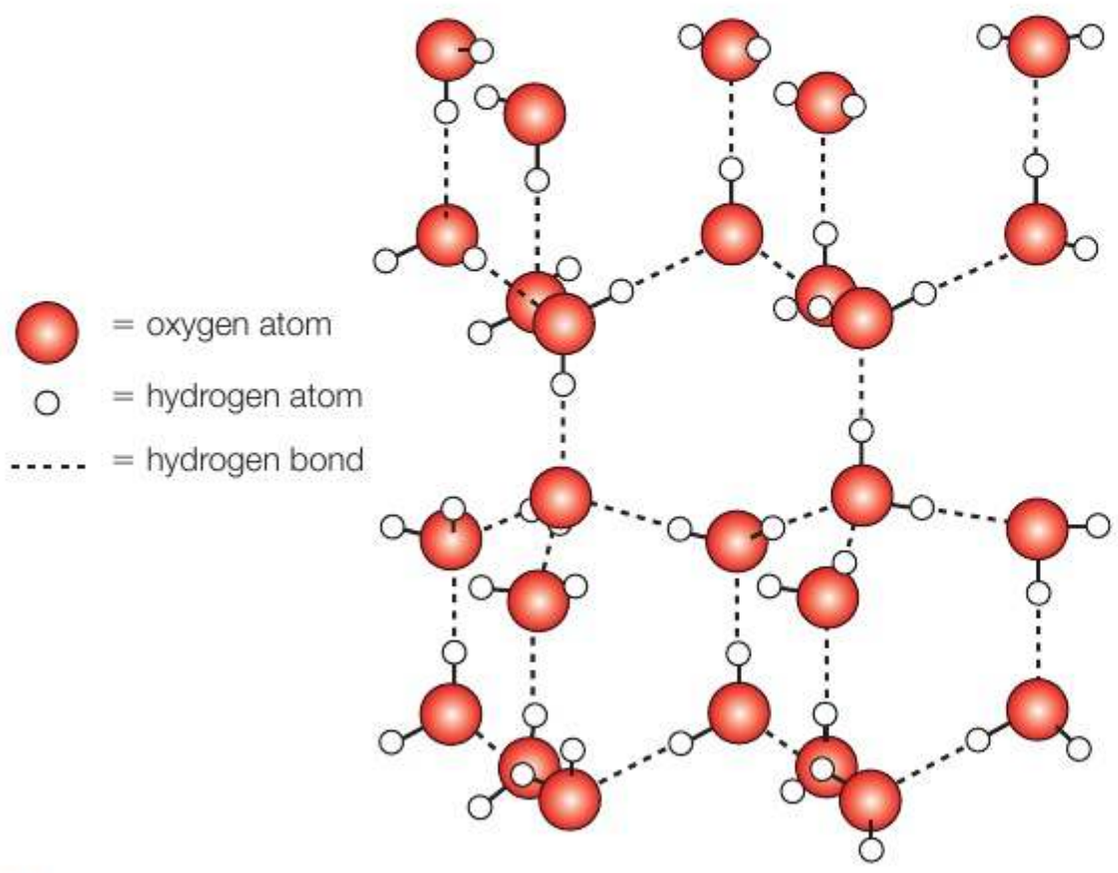
1 HF forms two hydrogen bonds per molecule, whereas water molecules can form up to four hydrogen bonds per molecule. This means that the hydrogen bonding is much more extensive in water.
2 Not all of the hydrogen bonds in HF are broken on vaporisation, since HF is substantially polymerised, even in the gas phase.

Ammonia has the lowest boiling temperature of the three compounds. The amount of hydrogen bonding in ammonia is limited by the fact that each nitrogen atom only has one lone pair. In a group of ammonia molecules, there are not enough lone pairs to go around to satisfy all the hydrogen atoms. This means that, on average, each ammonia molecule can form one hydrogen bond using its lone pair and one involving one of its $\delta+$ hydrogen atoms. The other hydrogen atoms are 'wasted'. The reason for the relatively low boiling temperature of ammonia is presumably the low strength of the hydrogen bonding between its molecules.

DENSITY OF ICE

As we mentioned above, water has another unusual property: the density of the solid (ice) is less than the density of the liquid at 0 °C.

The molecules in ice are arranged in rings of six, held together by hydrogen bonds (**fig D**).



▲ **fig D** Hydrogen bonding in ice.

The structure creates large areas of open space inside the rings. When ice melts, the ring structure is destroyed and the average distance between the molecules decreases, causing an increase in density.

CHOOSING SUITABLE SOLVENTS

For a substance to dissolve, the following two conditions must be met.

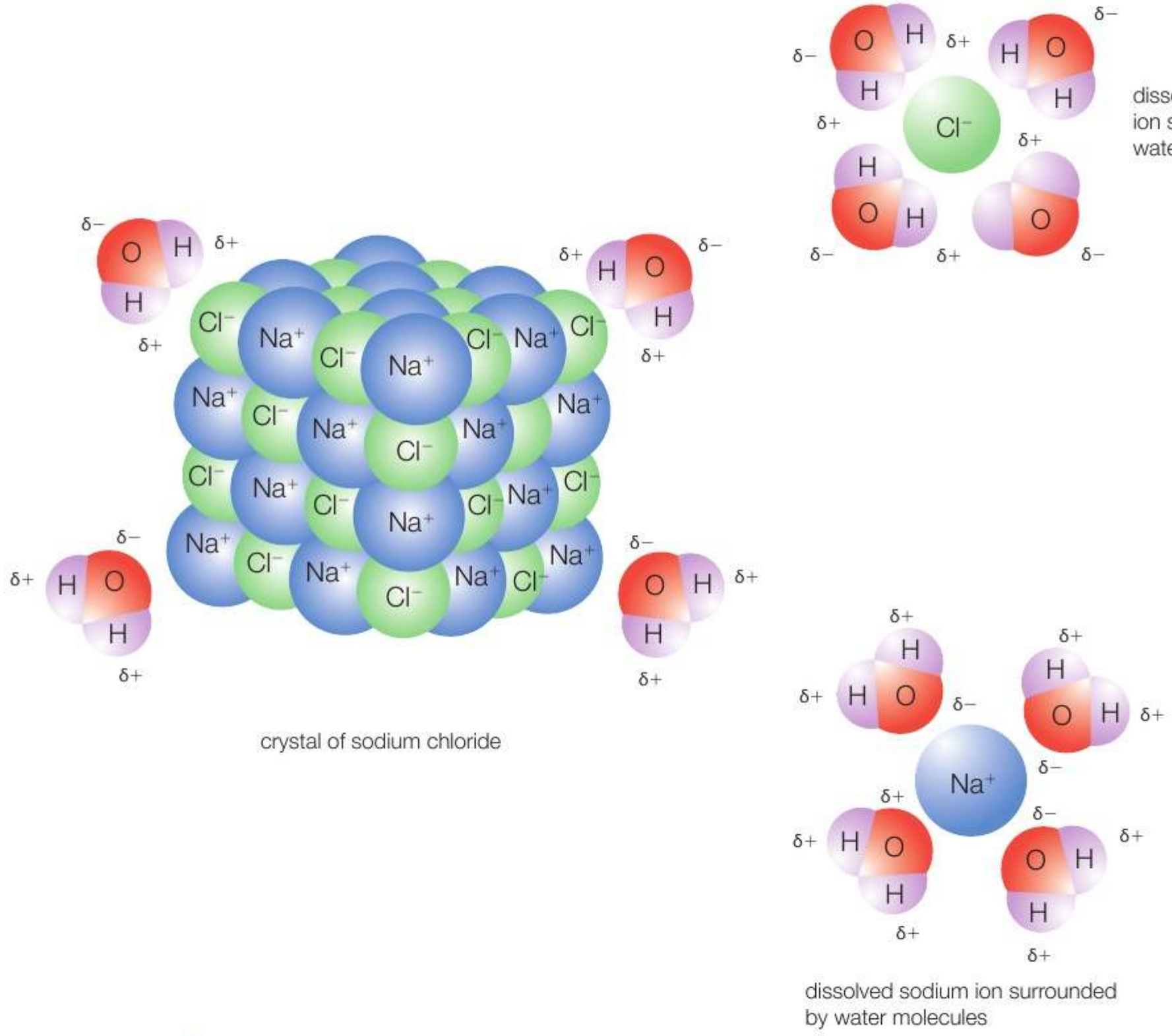
1 The solute particles must be separated from each other and then become surrounded by solvent particles.

2 The forces of attraction between the solute and solvent particles must be strong enough to overcome the solvent–solvent forces and the solute–solute forces.

DISSOLVING IONIC SOLIDS

Many ionic solids dissolve in water. The energy required to separate the ions in the solid is either completely, or partially, supplied by the hydration of the ions.

Fig E shows the process of dissolving for sodium chloride.



▲ **fig E** Dissolving of sodium chloride.

DID YOU KNOW?
This explanation is a simplification of the processes that occur. A more detailed explanation of the dissolving of ionic solids in water will be discussed in Topic 12 (Book 2: IAL).

The $\delta-$ ends of the water molecules attract the sodium ions sufficiently to remove them from the lattice. The sodium ions then become surrounded by water molecules, as shown. The interaction between the sodium ions and the water molecules is called an ion–dipole interaction.

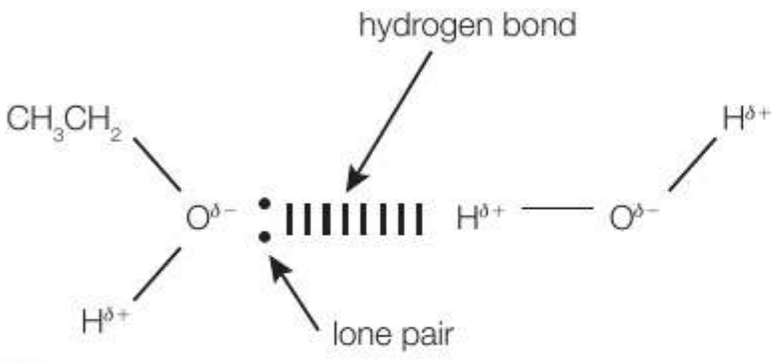
The $\delta+$ end of the water molecules attracts the chloride ions. Once they are in solution, the chloride ions become surrounded by water molecules, as shown. The chloride ions are hydrogen bonded to the water molecules.

The above processes are known as 'hydration' and the energy released is known as the 'hydration energy'.

COMPOUNDS THAT CAN FORM HYDROGEN BONDS WITH WATER

Alcohols contain an –O—H group and can therefore form hydrogen bonds with water.

Fig F shows a molecule of ethanol forming a hydrogen bond to water:



▲ **fig F** Hydrogen bonding between ethanol and water.

Ethanol and water mix in all proportions. The hydrogen bonding between the ethanol and water molecules is similar in strength to the hydrogen bonding in pure ethanol and in pure water.

The solubility of alcohols in water decreases with increasing hydrocarbon chain length as London forces predominate between the alcohol molecules.

COMPOUNDS THAT CANNOT FORM HYDROGEN BONDS WITH WATER

Non-polar molecules such as the alkanes do not dissolve in water. The attraction between the alkane molecules and water molecules is not sufficiently strong to disrupt the hydrogen bonded system between the water molecules.

Many polar molecules also have limited solubility in water. This is because they either do not form hydrogen bonds with water, or the hydrogen bonds they do form are weak compared with the hydrogen bonds in water.

Ethoxyethane, $CH_3CH_2OCH_2CH_3$, is polar (dipole moment = 1.15 D) and yet is almost totally immiscible with water. The forces of attraction between ethoxyethane and water molecules are not large enough to replace the relatively strong hydrogen bonding between the water molecules.

Halogenoalkanes (see **Topic 10**) are also not very soluble in water for similar reasons. They are much more soluble in ethanol, and this is why some reactions of halogenoalkanes are carried out in a medium of aqueous ethanol.

NON-AQUEOUS SOLVENTS

A general rule is that 'like dissolves like'. In other words, substances that are very similar dissolve each other. If you are searching for a solvent for a non-polar substance, or for a substance that has a substantial non-polar part to its molecule, then liquids that contain similar molecules are often the answer.

For example, alkanes are soluble in one another. Crude oil is a complex mixture of alkanes dissolved in each other.

Non-polar bromine dissolves readily in non-polar hexane (C_6H_{14}), and this solution is sometimes used to test for unsaturation in molecules. It is more convenient to use than bromine water, since the molecules being tested will also be soluble in hexane, whereas they are likely to be insoluble in water.

LEARNING TIP

The situation is complicated by the fact that molecules such as bromine both dissolve in, *and* react with, water. This type of reaction will be considered further in **Topic 8**.

CHECKPOINT

SKILLS REASONING

1. The hydrides of Group 4 exist as tetrahedral molecules.
Explain the trend in their boiling temperatures:

CH_4 (112 K) SiH_4 (161 K) GeH_4 (185 K) SnH_4 (221 K)

2. Suggest why water has a relatively high surface tension for a molecule of such low molecular mass.

3. A propanone molecule (CH_3COCH_3) has 32 electrons. A butane molecule ($CH_3CH_2CH_2CH_3$) has 34 electrons. Explain why propanone has a much higher boiling temperature than butane.

4. Suggest why magnesium chloride is soluble in water even though the energy required to break up the lattice is 2494 kJ mol^{-1}.